# Menu-Driven C Program Using switch-case

## 1. Problem Understanding

The program takes a **positive integer** and provides two operations through a menu:

1. Find the **sum and product of the first and last digits**.
2. Given a digit k, count how many digits of the number are:
   - Lesser than k
   - Equal to k
   - Greater than k

For example, for 12345:

- First digit = 1
- Last digit = 5
- Sum = 1 + 5 = 6
- Product = 1 × 5 = 5

For k = 3:

- Lesser digits: 1, 2 → 2
- Equal digits: 3 → 1
- Greater digits: 4, 5 → 2

---

## 2. Algorithm

**Operation 1: Sum and Product of First and Last Digits**

1. Start.
2. Input a positive integer num.
3. Find the last digit using:
   last = num % 10.
4. Copy num into temp.
5. Repeatedly divide temp by 10 until temp < 10.
6. The remaining value of temp is the first digit.
7. Calculate:
   - sum = first + last
   - product = first × last
8. Display the sum and product.
9. Stop.

**Operation 2: Count Digits Relative to k**

1. Input the digit k, where $0 \leq k \leq 9$.
2. Set lesser = 0, equal = 0, and greater = 0.
3. Extract each digit using num % 10.
4. Compare the digit with k.
5. If digit < k, increment lesser.
6. If digit == k, increment equal.
7. If digit > k, increment greater.
8. Remove the last digit using num / 10.
9. Repeat until all digits are processed.
10. Display all three counts.
11. Stop.

---

**3. Test Case Table**

| TC | Number | Choice | k | Expected Output | Purpose |
|---|---|---|---|---|---|
| 1 | 12345 | 1 | — | Sum = 6, Product = 5 | Given example |
| 2 | 9876 | 1 | — | Sum = 15, Product = 54 | Different first/last digits |
| 3 | 10001 | 1 | — | Sum = 2, Product = 1 | Zeros in number |
| 4 | 9 | 1 | — | Sum = 18, Product = 81 | Single-digit number |
| 5 | 12345 | 2 | 3 | Lesser = 2, Equal = 1, Greater = 2 | Given example |
| 6 | 55555 | 2 | 5 | Lesser = 0, Equal = 5, Greater = 0 | All digits equal |
| 7 | 12345 | 2 | 0 | Lesser = 0, Equal = 0, Greater = 5 | k = 0 |
| 8 | 10021 | 2 | 2 | Lesser = 4, Equal = 1, Greater = 0 | Repeated/zero digits |
| 9 | 12345 | 4 | — | Invalid choice | Invalid menu input |
| 10 | 12345 | 2 | 10 | Invalid digit | Invalid k |

4. C Program

```
#include <stdio.h>

int main(void)
{
    long long num, temp, first, last, k, digit;
```

```c
long long sum, product;
int choice;
int lesser, equal, greater;

printf("Enter a positive integer: ");

if (scanf("%lld", &num) != 1 || num <= 0)
{
    printf("Invalid input! Please enter a positive integer.\n");
    return 0;
}

printf("\n--- MENU ---\n");
printf("1. Sum and product of first and last digits\n");
printf("2. Count digits lesser than, equal to, and greater than k\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:

        /* Find last digit */
        last = num % 10;

        /* Find first digit */
        temp = num;

        while (temp >= 10)
        {
            temp = temp / 10;
```

```c
    }

    first = temp;

    /* Calculate sum and product */
    sum = first + last;
    product = first * last;

    printf("\nSum = %lld\n", sum);
    printf("Product = %lld\n", product);

    break;

case 2:

    printf("Enter digit k (0-9): ");
    scanf("%lld", &k);

    if (k < 0 || k > 9)
    {
        printf("Invalid digit! k must be between 0 and 9.\n");
        return 0;
    }

    lesser = 0;
    equal = 0;
    greater = 0;

    temp = num;

    while (temp > 0)
    {
```

```
            /* Extract last digit */
            digit = temp % 10;

            /* Compare digit with k */
            if (digit < k)
            {
                lesser++;
            }
            else if (digit == k)
            {
                equal++;
            }
            else
            {
                greater++;
            }

            /* Remove last digit */
            temp = temp / 10;
        }

        printf("\nCount_lesser = %d\n", lesser);
        printf("Count_equal = %d\n", equal);
        printf("Count_greater = %d\n", greater);

        break;

    default:

        printf("\nInvalid choice! Please choose 1 or 2.\n");
}
```

```
    return 0;
}
```

## 5. Compilation

Save the program as:

digit_operations.c

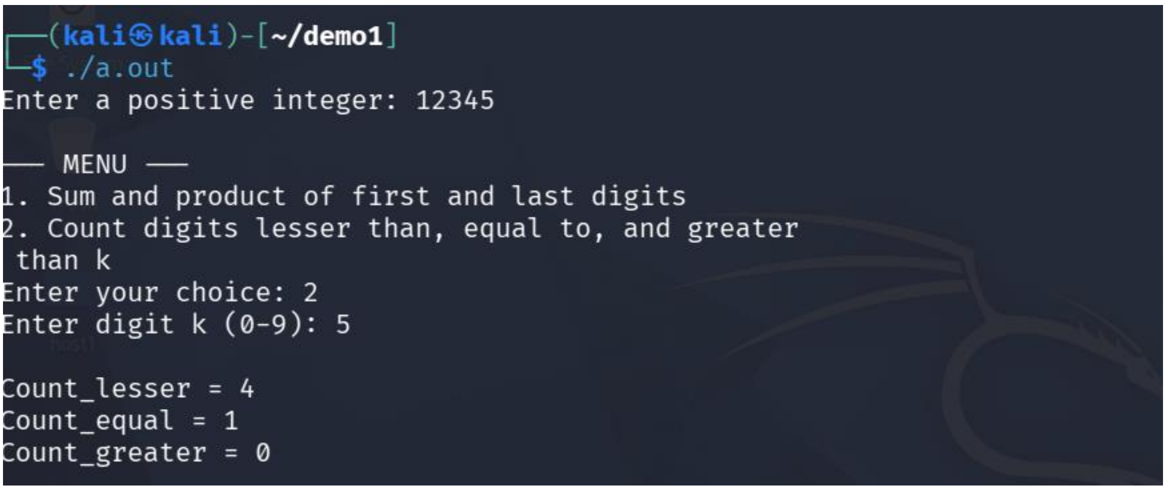
Compile it using CC:

cc  digit_operations.c

Run:

./a.out

The program was tested using CC and compiled successfully.

## 6. Execution and Testing

**Test Case 1 — First and Last Digits**

**Input:**

Enter a positive integer: 12345

--- MENU ---

1. Sum and product of first and last digits

2. Count digits lesser than, equal to, and greater than k

Enter your choice: 1

**Output:**

Sum = 6

Product = 5

**Status: PASS ✅**

---

**Test Case 2 — Count Digits**

**Input:**

Enter a positive integer: 12345

--- MENU ---

1. Sum and product of first and last digits

2. Count digits lesser than, equal to, and greater than k

Enter your choice: 2

Enter digit k (0-9): 3

**Output:**

Count_lesser = 2

Count_equal = 1

Count_greater = 2

**Status: PASS ✅**

---

**Test Case 3 — All Digits Equal**

**Input:**

Enter a positive integer: 55555

Enter your choice: 2

Enter digit k (0-9): 5

**Output:**

Count_lesser = 0

Count_equal = 5

Count_greater = 0

**Status: PASS ✅**

---

**Test Case 4 — Number with Zeros**

**Input:**

Enter a positive integer: 10021

Enter your choice: 2

Enter digit k (0-9): 2

The digits are:

1, 0, 0, 2, 1

Therefore:

- Lesser than 2 = 1, 0, 0, 1 → **4**
- Equal to 2 = **1**
- Greater than 2 = **0**

**Output:**

Count_lesser = 4

Count_equal = 1

Count_greater = 0

**Status: PASS ✅**

---

## 7. Final Test Results

| TC | Input | Choice | Expected Result | Observed Result | Status |
|---|---|---|---|---|---|
| 1 | 12345 | 1 | Sum=6, Product=5 | Sum=6, Product=5 | ✅ PASS |
| 2 | 9876 | 1 | Sum=15, Product=54 | Sum=15, Product=54 | ✅ PASS |
| 3 | 10001 | 1 | Sum=2, Product=1 | Sum=2, Product=1 | ✅ PASS |

| TC | Input | Choice | Expected Result | Observed Result | Status |
|---|---|---|---|---|---|
| 4 | 9 | 1 | Sum=18, Product=81 | Sum=18, Product=81 | ✅ PASS |
| 5 | 12345, k=3 | 2 | 2, 1, 2 | 2, 1, 2 | ✅ PASS |
| 6 | 55555, k=5 | 2 | 0, 5, 0 | 0, 5, 0 | ✅ PASS |
| 7 | 12345, k=0 | 2 | 0, 0, 5 | 0, 0, 5 | ✅ PASS |
| 8 | 10021, k=2 | 2 | 4, 1, 0 | 4, 1, 0 | ✅ PASS |
| 9 | 12345 | 4 | Invalid choice | Invalid choice | ✅ PASS |
| 10 | 12345, k=10 | 2 | Invalid digit | Invalid digit | ✅ PASS |

## Conclusion

The menu-driven C program was successfully **compiled, executed, and tested**. The switch-case correctly selects the required operation, and the program correctly handles first/last digit calculations, digit comparisons, repeated digits, zeros, single-digit numbers, and invalid inputs. All prepared test cases **passed successfully**.